IS AN EXPENSIVE AGENT ALWAYS THE WAY TO GO?

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1 PROBLEM DEFINITION AND MOTIVATION

Nowadays, people keep talking about AI agents and their capabilities. They say the future is agentic, and you should try incorporating more and more agents into your life. However, much of the work routed through these expensive frontier models is just routine decisions that users could instead use a cheap calibrated decision model to take actions. The model would have to be able to tell apart these routine decisions from actions requiring more reasoning.

We propose a model which uses the new Jev model or its open source counterpart Kev to take these quick routing decisions, with an agent we would create, capable of selecting knowledge base tools, gathering additional evidence, and revising a decision. We would use datasets such as Banking77, a dataset composed of online banking queries annotated with their corresponding intents, and CLINC150, an evaluation dataset for intent classification and out-of-scope prediction. Then if time allows, we would create our own dataset from which we could be confident that the models wouldn't have had training on.

2 COMPARISON WITH EXISTING WORK

Chen et al. (2023) propose FrugalGPT, which queries models in sequence with increasing cost and only accept a models answer when a learned reliability score exceeds a learned threshold. The caveat is that both the scorer and the threshold are learned on labeled data which implicitly assume that the deployment examples will resemble the labeled training data and whether the model is reliable on unfamiliar data is hard to state. RouteLLM (Ong et al., 2024) use a different approach, learning from preference data whether to choose a stronger or weaker model, with it objective being relative response quality and not a bound on the errors the cheaper model is allowed to make. Most recently, Calibration-Aware Uncertainty Cascades(Zhang et al., 2026) calibrate confidence to decise wehen to quit and when to escale. Different from our project their escalation is a single model call, instead of a trigger to an agent.

The method of accepting a models answer only when it is certain and abstaining from uncertain answers is a form of selective classification (Geifman & El-Yaniv, 2017), which are usually summarized by a risk–coverage curve. They obviously depend on confidence scores being well calibrated, which modern neural networks often are not without post-hoc correction (Guo et al., 2017).

Our project differs from the mentioned work in three ways. Like mentioned above the fallback is not a larger model but an agent that retrieves knowledge-base evidence and may rethink a decision and use tools, so escalation is more expensive and variable in cost. Instead of optimizing the average trade-off between accuracy and cost, the automatically accepted decisions are under a fixed error budget and we measure the coverage possible under it. and then measure the coverage achievable under it. Lastly, we test whether thresholds transfer between datasets and whether it can handle outlier inputs. We analyze errors for which the model is highly confident on against provided labels.

We intend to contribute empirical evidence for when a fast decision layer like Jev/Kev (Almeida, 2026; Palmer, 2026) can safely reduce agent use and when it cannot.

3 Next Steps and Direction

Next steps would be to start implementing this system. We would use a Qwen model/one of OpenAI's Astra models to be the escalating agent, and start with task specific prompts. Then we would implement the quick decision stage using the Jev/Kev models, calibrating their confidence scores if needed. We'd choose a confidence threshold on a development set to target at most 5% error on accepted decisions, then evaluate on a held out test set. We'd test the performance, both on cost and time, and do a Pareto plot to compare accuracy vs cost, alongside a risk coverage plot. We'd compare this against each stage alone and random escalation at the same rate. We would know this works if the quick decision models, Jev/Kev, reduce cost and ideally time whilst meeting the error target and keeping overall accuracy close to the agent only baseline. We'd also look at high confidence mistakes and whether the threshold works across datasets.

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4 AI-USE DISCLOSURE STATEMENT

No AI was used to generate the text in this report besides the references, where we got ChatGPT Sol 6.1 with High reasoning to find the proper references. The same model was used to bounce off ideas and formalize thoughts, but all the writing was done by us. No coding has taken place yet.